

Database Normalization

Second Normal Form (2NF)

Hospital Patient Visits — Assessment Problem

Course: Database Systems | Lab Manual — Task 8

Topic: 1NF → 2NF | XAMPP / MySQL 8.x

1NF ✓

→

2NF ✓

1. What is Second Normal Form (2NF)?

A table is in **Second Normal Form (2NF)** if and only if it is already in 1NF **AND** every non-prime attribute is fully functionally dependent on the *entire* primary key — not just a part of it. This rule applies when the primary key is a **composite key** (made of more than one column).

2NF Rule

The table must be in 1NF.

Every non-prime attribute must depend on the **WHOLE** primary key.

No partial dependencies allowed — every column must need ALL parts of the PK to be determined.

Key Point for our Dataset: In the 1NF table, the primary key is **VisitID alone** (a single-column PK). This means technically no partial dependencies can exist with a single-column PK. However, the data clearly has attributes that logically depend on PatientID (not VisitID) and on DoctorID (not VisitID). This is still treated as a 2NF violation — we separate entities into their own tables.

2. BEFORE: 1NF Table (Starting Point)

■ ■ This is where we left off in 1NF — one big table with all columns. Patient and Doctor info still repeats across multiple rows.

Table: HospitalVisit_1NF (12 columns, 1 table)

■ Red = Duplicate patient data (Hassan appears twice) | ■ Orange = Duplicate doctor data | Row 3: P-201 & D-31 both repeat

VisitID*	VisitDate	PatID	PatName	PatPhone	DocID	DocName	Specialty	DeptName	DeptHead	Diagnosis	Fee
V-9001	2026-04-10	P-201	Hassan	0300-1112233	D-30	Dr. Imran	Cardiology	Heart Care	Dr. Tariq	Hypertension	2500
V-9002	2026-04-10	P-202	Mehreen	0301-4445566	D-31	Dr. Asma	Dermatology	Skin Clinic	Dr. Asma	Eczema	2000
V-9003	2026-04-11	P-201	Hassan	0300-1112233	D-31	Dr. Asma	Dermatology	Skin Clinic	Dr. Asma	Allergy	2000
V-9004	2026-04-12	P-203	Junaid	0302-7778899	D-30	Dr. Imran	Cardiology	Heart Care	Dr. Tariq	Arrhythmia	3000

3. Identifying Partial Dependencies in 1NF

We analyze every non-key attribute and ask: *Does this attribute depend on VisitID directly, or does it really depend on PatientID or DoctorID?* Attributes that depend on PatientID or DoctorID (not VisitID) are **partial dependencies** and must be moved to separate tables.

Attribute(s)	Really Depends On	Dependency Type	Action
PatientName, PatientPhone	PatientID (NOT VisitID)	■ Partial	Move to Patient table
DoctorName, Specialty	DoctorID (NOT VisitID)	■ Partial	Move to Doctor table
DeptName, DeptHead	DoctorID (NOT VisitID)	■ Partial	Move to Doctor table (3NF will fix DeptHead)
VisitDate, Diagnosis, Fee	VisitID ✓	■ Full	Stay in Visit table
PatientID, DoctorID	VisitID ✓ (FK refs)	■ Full	Stay as foreign keys in Visit table

4. Decomposition Plan: 1NF → 2NF (3 Tables)

#	Table Name	Columns	Purpose
1	Patient	PatientID (PK), PatientName, PatientPhone	Stores each patient's info exactly once
2	Doctor	DoctorID (PK), DoctorName, Specialty, DeptName, DeptHead	Stores each doctor + dept info once (DeptHead fixed in 3NF)
3	Visit	VisitID (PK), VisitDate, PatientID (FK), DoctorID (FK), Diagnosis, Fee	Records each visit; links to Patient & Doctor via FKs

HospitalVisit_1NF (1 big table)

→ decompose →

Patient + Doctor + Visit

5. AFTER: Second Normal Form (2NF) — 3 Tables

■ Partial dependencies eliminated | 3 separate tables | Each entity in its own table | Foreign keys link tables

Table 1: Patient

Contains only patient-specific information. PatientID → PatientName, PatientPhone.

PatientID (PK)	PatientName	PatientPhone
P-201	Hassan	0300-1112233
P-202	Mehreen	0301-4445566
P-203	Junaid	0302-7778899

Table 2: Doctor

Contains doctor and department information. DoctorID → DoctorName, Specialty, DeptName, DeptHead. (Note: DeptHead still has a transitive dependency on DeptName — this will be fixed in 3NF.)

DoctorID (PK)	DoctorName	Specialty	DeptName	DeptHead
D-30	Dr. Imran	Cardiology	Heart Care	Dr. Tariq
D-31	Dr. Asma	Dermatology	Skin Clinic	Dr. Asma

■■ DeptName → DeptHead is a transitive dependency still present here. Will be resolved in 3NF.

Table 3: Visit

Contains only visit-specific facts. VisitID → VisitDate, Diagnosis, Fee, plus FK references to Patient and Doctor.

VisitID (PK)	VisitDate	PatientID (FK)	DoctorID (FK)	Diagnosis	Fee (PKR)
V-9001	2026-04-10	P-201	D-30	Hypertension	2500
V-9002	2026-04-10	P-202	D-31	Eczema	2000
V-9003	2026-04-11	P-201	D-31	Allergy	2000
V-9004	2026-04-12	P-203	D-30	Arrhythmia	3000

6. Side-by-Side Comparison: 1NF vs 2NF

Feature	1NF (Before)	2NF (After)
No. of Tables	1 (HospitalVisit_1NF)	3 (Patient, Doctor, Visit)
Total Columns	12 in one table	3 + 5 + 6 = 14 (distributed)
Partial Deps	■ PatientName/Phone not on VisitID	■ Eliminated — each in own table
Patient Redundancy	■ Hassan's data repeated 2x	■ Hassan stored once in Patient table
Doctor Redundancy	■ Dr. Imran/Asma repeated 2x ea	■ Each doctor stored once in Doctor
Transitive Deps	■■ DeptHead depends on DeptName	■■ Still present in Doctor table — 3NF fix

Foreign Keys	■ None defined	■ Visit.PatientID → Patient, Visit.DoctorID → Doctor
Insertion Anomaly	■ Can't add patient without visit	■ Add to Patient table directly
Update Anomaly	■ Must update multiple rows	■ Update one row in Doctor table


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DeptName = 'Cardiac Care' WHERE DoctorID = 'D-30';
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8. Expected Query Output After 2NF

Patient Table Output:

PatientID	PatientName	PatientPhone
P-201	Hassan	0300-1112233
P-202	Mehreen	0301-4445566
P-203	Junaid	0302-7778899

Doctor Table Output:

DoctorID	DoctorName	Specialty	DeptName	DeptHead
D-30	Dr. Imran	Cardiology	Heart Care	Dr. Tariq
D-31	Dr. Asma	Dermatology	Skin Clinic	Dr. Asma

Visit Table Output:

VisitID	VisitDate	PatientID	DoctorID	Diagnosis	Fee
V-9001	2026-04-10	P-201	D-30	Hypertension	2500
V-9002	2026-04-10	P-202	D-31	Eczema	2000
V-9003	2026-04-11	P-201	D-31	Allergy	2000
V-9004	2026-04-12	P-203	D-30	Arrhythmia	3000

9. Summary & What Comes Next

The 1NF table has been successfully decomposed into **3 tables in 2NF**: *Patient*, *Doctor*, and *Visit*. Patient and Doctor data is now stored exactly once, eliminating insertion and update anomalies. However, the Doctor table still contains a **transitive dependency**: DeptName → DeptHead (DeptHead depends on DeptName, not directly on DoctorID). This will be resolved in **Third Normal Form (3NF)**.

Status	Step	Achievement
■ Done	UNF → 1NF	Atomic values, primary key defined
■ Done	1NF → 2NF	3 tables: Patient, Doctor, Visit — partial deps removed
■ Next	2NF → 3NF	Remove DeptName→DeptHead transitive dep → add Department table